CSE470: Software Engineering

Requirement Engineering

What is a Requirement?

It may range from a high-level abstract statement of a service or of a system constraint to a detailed mathematical functional specification.

This is inevitable as requirements may serve a dual function

- May be the basis for a bid for a contract – therefore must be open to interpretation;
- May be the basis for the contract itself – therefore must be defined in detail;
- Both these statements may be called requirements.

Types of Requirement

Functional requirements

- Statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.
- May state what the system should not do.

Non-functional requirements

- Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.
- Often apply to the system as a whole rather than individual features or services.

Functional Requirements

- Describe functionality or system services.
- Depend on the type of software, expected users and the type of system where the software is used.
- Functional user requirements may be high-level statements of what the system should do.
- Functional system requirements should describe the system services in detail.

Non-Functional Requirements

- These define system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc.

- Process requirements may also be specified mandating a particular IDE, programming language or development method.

- Non-functional requirements may be more critical than functional requirements. If these are not met, the system may be useless.

Non-Functional Classifications

Product requirements

- Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

Organisational requirements

- Requirements which are a consequence of organisational policies and procedures e.g. process standards used, implementation requirements, etc. (meeting budget limitations, company guidelines on data handling, recommended CASE tools)

External requirements

- Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

Functional VS Non-Functional Requirements

Sl. No	Functional Requirement	Non-functional Requirement
1	Defines all the services or functions required by the customer that must be provided by the system	Defines system properties and constraints e.g. reliability, response time and storage requirements. Constraints are I/O device capability, system representations, etc
2	It describes what the software should do	It does not describe what the software will do, but how the software will do it
3	Related to business. For example: Calculation of order value by Sales Department or gross pay by the Payroll Department	Related to improving the performance of the business. For example: checking the level of security. An operator should be allowed to view only my name and personal identification code
4	Functional requirements are easy to test	Non-functional requirements are difficult to test
5	Related to the individual system features	Related to the system as a whole
6	Failure to meet the individual functional requirement may degrade the system	Failure to meet a non-functional requirement may make the whole system unusable

Examples

Functional requirement
1. A user shall be able to search the appointments lists for all clinics.
2. The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
3. Each staff member using the system shall be uniquely identified by his or her eight-digit employee number.

Product requirement
The MHC-PMS shall be available to all clinics during normal working hours (Mon–Fri, 0830–17.30). Downtime within normal working hours shall not exceed five seconds in any one day.

Organizational requirement
Users of the MHC-PMS system shall authenticate themselves using their health authority identity card.

External requirement
The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

A few things to note

- You must write your requirements in full sentences
- While writing functional requirements, it should start with: “User should be able to”, “System should allow user to” etc.
- While writing your non functional requirements, you can use “The system should have”, “System should be able to”, “The system should implement”.

Here “user” means the specific user of that system like students, drivers etc.

Class Task

You have been hired to develop a smart parking management system for a multi-story shopping mall. The mall management wants to automate parking space allocation, payment, and security monitoring to improve customer experience and reduce manual work.

The system should allow drivers to check space availability in real-time, reserve a spot before arriving, and make cashless payments. The system should also support security features like automatic license plate recognition and emergency alert buttons. The system should be able to handle large amounts of user load and ensure near perfect uptime. The system should also implement encryption mechanisms to safeguard user data.

The mall wants to first launch a basic set of features as a pilot on a single floor, then gradually add advanced features like dynamic pricing, loyalty programs, and integration with shopping apps, depending on user feedback.

- **Determine the Functional and Non-Functional Requirements**

Class Task - Answer

Functional Requirements

1. The system shall allow drivers to view real-time parking space availability on a mobile app and kiosk screens.
2. The system shall allow drivers to reserve a parking spot in advance using a mobile application.
3. The system shall enable drivers to make cashless payments (credit card, mobile wallet, etc.).
4. The system shall support automatic license plate recognition to verify entry and exit.
5. The system shall allow drivers to trigger an emergency alert from kiosks or mobile apps if they face security threats.

Non-Functional Requirements

1. The system shall be able to handle at least 500 simultaneous users without performance degradation.
2. The system shall have an availability of 99.9% uptime to ensure continuous service.
3. The system shall encrypt all user data to ensure user data privacy